

# **FITBUDDY-AI FITNESS PLAN GENERATOR USING GEMINI MODELS**

## **Phase 5: Project Development Phase**

This is the main implementation phase where the FitBuddy application is developed.

### **5.1 AI Model Integration**

Google Gemini models are integrated into the application for:

- Workout generation.
- Nutrition/recovery tips.
- Workout plan modification.

### **5.2 Workout Plan Generation**

The `generate_workout_gemini()` function processes user information and sends an appropriate prompt to the Gemini model.

The generated plan contains:

- 7-day schedule.
- Warm-up instructions.
- Exercises.
- Sets and repetitions.
- Rest periods.
- Cooldown/recovery guidance.

### **5.3 Nutrition Tip Generation**

The `generate_nutrition_tip_with_flash()` function generates concise nutrition or recovery advice based on the user's fitness goal.

Examples include:

- Hydration guidance.
- Protein-related suggestions.
- Recovery practices.
- General healthy eating recommendations.

## **5.4 Feedback-Based Plan Updating**

The `update_workout_plan()` function receives:

- The existing workout plan.
- User feedback.

The AI then generates a revised workout plan according to the requested changes.

## **5.5 Backend Development**

FastAPI routes are implemented for:

- `/`
- `/generate-workout`
- `/submit-feedback`
- `/view-all-users`

Form inputs are processed using FastAPI's `Form(...)`, while Pydantic models are used for structured input validation.

## **5.6 Database Development**

SQLAlchemy and SQLite are used to implement functions such as:

- `save_user()`
- `save_plan()`
- `update_plan()`
- `get_original_plan()`
- `get_user()`
- `get_all_users()`
- `get_all_plans()`

## **5.7 Frontend Development**

Jinja2 templates are developed to:

- Collect user information.
- Display generated workout plans.
- Display nutrition tips.
- Collect feedback.
- Display updated plans.
- Provide an administrative user view.

### **Outcome:**

A functional FitBuddy application integrating AI, backend services, database storage, and frontend interfaces.